

Greedy Classes Need Not Be Linear

The question in arXiv:1303.4972 was answered by arXiv:0911.4900

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Status: literature already answered (negative). No new theorem is claimed. The literal universal question has an explicit counterexample in the cited earlier paper of Garrigós, Hernández, and de Natividade. Choosing its exponents $p = 2$ and $r = 1$ keeps the example inside the Banach-space and lattice-unconditional setting of arXiv:1303.4972.

1 The source question

For a normalized basis (e_n) in a Banach space, let $\gamma_N(x)$ be the largest error among thresholding greedy approximants of order N (the maximum only matters when coefficients tie). Wojtaszczyk defines the greedy classes by replacing the best N -term error in the usual approximation-space quasi-norm by $\gamma_N(x)$. For the endpoint parameter,

$$\mathcal{G}_\infty^\alpha = \left\{ x : \|x\| + \sup_{N \geq 1} N^\alpha \gamma_N(x) < \infty \right\}.$$

The paper then says that it is not known whether $\mathcal{G}_q^\alpha$ is a linear space. Read as the natural universal question over the paper's fixed but otherwise arbitrary unconditional basis, the answer is no.

$$\sigma_N(x) = \inf \left\{ \left\| x - \sum_{n \in \Lambda} b_n x_n \right\| : \#\Lambda = N \text{ and } b_n \text{'s are arbitrary} \right\} \quad (11)$$

2. Greedy classes $\mathcal{G}_q^\alpha$ are defined in the same way but we replace $\sigma_N(x)$ by error of a greedy approximation which is defined as $\gamma_N(x) = \max \|x - \mathcal{G}_N(x)\|$. The maximum is taken over all $\mathcal{G}_N(x)$'s in case it is not uniquely defined.

It is well known that $\mathcal{A}_q^\alpha$ are quasi-Banach spaces with the quasi-norm $\|\cdot\|_{\mathcal{A}_q^\alpha}$. For the spaces $\mathcal{G}_q^\alpha$ the situation is not so clear—we do not know if it is a linear space. Clearly if the basis is greedy then $\sigma_N \sim \gamma_N$ and the spaces are equal. Also, since always $\sigma_N(x) \leq \gamma_N(x)$, we have $\mathcal{G}_q^\alpha \subset \mathcal{A}_q^\alpha$. The problem wether the equality $\mathcal{G}_q^\alpha = \mathcal{A}_q^\alpha$ characterise greedy bases was considered in [1]. Actually it turned out to be quite difficult so the authors considered the problem of equivalence of quantities $\|x\|_{\mathcal{A}_q^\alpha}$ and $\|x\|_{\mathcal{G}_q^\alpha}$. Let us say that *greedy*

Figure 1: Wojtaszczyk, arXiv:1303.4972, PDF page 5.

2 The earlier counterexample

Section 7.2 of Garrigós–Hernández–de Natividad (arXiv:0911.4900, published in 2012) takes

$$\mathbb{B} = \ell^p \oplus_{\ell^1} \ell^r, \quad 0 < r < p < \infty,$$

with its canonical basis. To match the later paper's hypotheses, take for example $p = 2$ and $r = 1$. Then $\mathbb{B}$ is a Banach space and the canonical basis is normalized and lattice unconditional.

Fix $\alpha > 0$, put

$$\beta = \alpha + \frac{1}{p}, \quad \gamma = \alpha + \frac{1}{r},$$

and define

$$x = ((k^{-\beta})_{k \geq 1}, 0), \quad y = (0, (j^{-\gamma})_{j \geq 1}).$$

Since $\beta p = \alpha p + 1$ and $\gamma r = \alpha r + 1$, integral comparison gives

$$\gamma_N(x) = \left(\sum_{k > N} k^{-\beta p} \right)^{1/p} \asymp N^{-\alpha}, \quad \gamma_N(y) = \left(\sum_{j > N} j^{-\gamma r} \right)^{1/r} \asymp N^{-\alpha}.$$

Consequently $x, y \in \mathcal{G}_\infty^\alpha$.

$$\frac{\gamma_N(x)}{h_\ell(p_N)} \gtrsim \frac{N^{-\alpha}}{N^{a\beta} (2^{N^b})^\beta} \approx \frac{N^{-\alpha}}{N^{a\beta}} = N^{b\gamma - a\beta} > N^{a\alpha} = \left(\frac{p_N}{q_N} \right)$$

which proves (7.1) in this case, so that we can apply Proposition 7.1 to conclude the result. The proof of *ii*) is similar with the same choice of p_N and q_N . $\square$

7.2. Non linearity of $\mathcal{G}_q^\alpha(\mathcal{B}, \mathbb{B})$. We conclude by showing with simple examples that $\mathcal{G}_q^\alpha(\mathcal{B}, \mathbb{B})$ may not even be a linear space when the basis $\mathcal{B}$ is not democratic.

Let $\mathbb{B} = \ell^p \oplus_{\ell^1} \ell^q$, $0 < q < p < \infty$; that is, $\mathbb{B}$ consists of pairs $(a, b) \in \ell^p \times \ell^q$, endowed with the quasi-norm $\|a\|_{\ell^p} + \|b\|_{\ell^q}$. We consider the canonical basis in $\mathbb{B}$.

Now, set $\beta = \alpha + \frac{1}{p}$ and $x = \{(k^{-\beta}, 0)\}_{k \in \mathbb{N}} \in \mathbb{B}$. For $N = 1, 2, 3, \dots$ we have

$$\gamma_N(x) = \left(\sum_{k > N} \frac{1}{k^{\beta p}} \right)^{1/p} \approx \left(\frac{1}{N^{\beta p - 1}} \right)^{1/p} = N^{-\alpha}.$$

Figure 2: Earlier paper, PDF page 21: start of Section 7.2.

3 Why the sum leaves the class

Here $r < p$, hence $\gamma > \beta$. Group the x -coefficients whose sizes lie between two consecutive y -coefficients:

$$A_j = \{k : j^{-\gamma} \leq k^{-\beta} < (j-1)^{-\gamma}\}.$$

Then $|A_j| \asymp j^{\gamma/\beta-1}$. At the greedy orders

$$N_J = J + \sum_{j=1}^J |A_j| \asymp J^{\gamma/\beta},$$

the retained coefficients are, up to harmless boundary ties, precisely the first J coefficients of y together with the x -coefficients in the first J blocks. Therefore

$$\begin{aligned} \gamma_{N_J}(x+y) &\asymp \left(\sum_{k > J^{\gamma/\beta}} k^{-\beta p} \right)^{1/p} + \left(\sum_{j > J} j^{-\gamma r} \right)^{1/r} \\ &\asymp J^{-\alpha\gamma/\beta} + J^{-\alpha} \asymp J^{-\alpha} \asymp N_J^{-\alpha\beta/\gamma}. \end{aligned}$$

It follows that

$$N_J^\alpha \gamma_{N_J}(x+y) \asymp N_J^{\alpha(1-\beta/\gamma)} \longrightarrow \infty,$$

so $x+y \notin \mathcal{G}_\infty^\alpha$. Thus $\mathcal{G}_\infty^\alpha$ is not a linear space, for every $\alpha > 0$, even for a normalized lattice-unconditional basis of a Banach space. This settles the literal universal question negatively. The earlier paper also states that a simple modification treats every finite class parameter, but that stronger assertion is not needed for the negative answer recorded here.

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This shows that $x \in \mathcal{G}_\infty^\alpha(\mathcal{B}, \mathbb{B})$. Similarly, if we let $\gamma = \alpha + \frac{1}{q}$, then $y = \{(0, j^{-\gamma})\}_{j \in \mathbb{N}}$ belongs to $\mathcal{G}_\infty^\alpha$. We will show, however, that $x+y \notin \mathcal{G}_\infty^\alpha$. In fact, we will find a subsequence N_J of natural numbers so that

$$\gamma_{N_J}(x+y) \approx \frac{1}{N_J^{\alpha\beta/\gamma}} \quad (7.8)$$

(notice that $\beta < \gamma$ since we chose $q < p$). To prove (7.8) let $A_1 = \{1\}$ and

$$A_j = \left\{ k \in \mathbb{N} : \frac{1}{j^\gamma} \leq \frac{1}{k^\beta} < \frac{1}{(j-1)^\gamma} \right\}, \quad j = 2, 3, \dots$$

The number of elements in A_j is

$$|A_j| \approx j^{\gamma/\beta} - (j-1)^{\gamma/\beta} \approx j^{\frac{\gamma}{\beta}-1}, \quad j = 1, 2, 3, \dots \quad (7.9)$$

For $J = 2, 3, 4, \dots$ let $N_J = \sum_{j=1}^J |A_j| + J$. From (7.9) we obtain

$$N_J \approx \sum_{j=1}^J j^{\frac{\gamma}{\beta}-1} + J \approx J^{\frac{\gamma}{\beta}} + J \approx J^{\frac{\gamma}{\beta}},$$

since $\gamma > \beta$. Thus,

$$\begin{aligned} \gamma_{N_J}(x+y) &\approx \left(\sum_{k > J^{\gamma/\beta}} k^{-\beta p} \right)^{1/p} + \left(\sum_{j > J} j^{-\gamma q} \right)^{1/q} \approx [(J^{\gamma/\beta})^{-\beta p+1}]^{1/p} + [J^{-\gamma q+1}]^{1/q} \\ &= J^{-\alpha\gamma/\beta} + J^{-\alpha} \approx J^{-\alpha} \approx (N_J)^{-\alpha\beta/\gamma}, \end{aligned}$$

Figure 3: Earlier paper, PDF page 22: the rate computation and conclusion.

4 Provenance and scope

The question was checked directly on page 5 of arXiv:1303.4972. The complete counterexample was checked on pages 21–22 of arXiv:0911.4900. The latter is G. Garrigós, E. Hernández, and M. de Natividade, *Democracy functions and optimal embeddings for approximation spaces*, Advances in Computational Mathematics 37 (2012), 255–283, DOI 10.1007/s10444-011-9197-0. It is cited as reference [1] by arXiv:1303.4972. This packet records an earlier-literature disposition only; it makes no novelty claim. If the source sentence was intended to ask about one particular basis constructed elsewhere in that paper rather than the general class just defined, the earlier example does not decide that narrower, unstated question.